
Procta

AI-proctored exams, built for India.

On-device AI proctoring with zero raw-video storage, native UPI and GST billing, and a published per-student price roughly an order of magnitude below the incumbents.

Investor briefing | July 2026 | Seeking pre-seed (~Rs. 4.8 lakh, 18-month runway)

THE PROBLEM

India runs more high-stakes exams than anywhere on earth - and the proctoring tools don't fit.

Coaching institutes and universities run millions of proctored exams a year, but the available platforms were built for someone else:

- **Priced in dollars.** US vendors (Honorlock, Proctortrack) charge per-attempt rates that convert to hundreds of rupees, plus FX risk.
- **Incumbents drifted away.** Mercer Mettl pivoted toward enterprise hiring assessments; education is no longer the core motion.
- **Heavy by design.** Continuous video upload means high bandwidth on budget student laptops and a large privacy surface under the DPDP Act.
- **Opaque pricing.** No competitor publishes a per-student rate, so every procurement conversation starts from zero.

Dollar-priced

incumbent per-attempt cost

Video-heavy

continuous upload by default

The gap is structural, not cosmetic: it is a cost-base and architecture mismatch that an India-built, on-device platform is positioned to close.

THE SOLUTION

Procta does the AI on the student's device, so the cost base and the privacy surface both collapse.

Face detection, gaze tracking, head-pose, cheat-object detection, voice activity and spoken-keyword spotting all run client-side on the student's CPU. The server only ever sees short JPEG snapshots of flagged moments - never a continuous video stream.

On-device ML pipeline (data-flow)



Raw camera frames never leave the student device - only short JPEG snapshots of flagged moments are uploaded.

A complete exam platform, not just a camera widget.

Live proctoring engine

- 8 detection signals: face, gaze, head-pose, YOLO cheat-objects, audio, behavioural correlation, wrong-person, multi-face.
- Spoken-keyword detection (Vosk en-IN + hi-IN) - unique among the compared vendors.
- Phone-camera secondary view via QR pairing, no separate app install.

Lockdown client

- Electron kiosk browser with process-integrity scan, auto-update, and offline answer resilience.

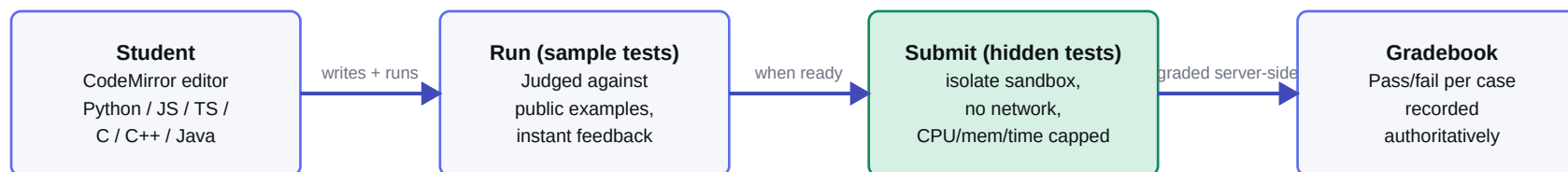
Teacher dashboard

- Live sessions, AI triage, forensics timeline, cluster review for bulk false-positive dismissal.
- Question bank, templates, groups, in-exam chat, onboarding wizard.

Grading and assessment

- Auto-graded MCQ, AI-suggested short-answer grading, composite risk score, AI session narrative.
- Inbuilt coding questions: server-side sandboxed execution judges Python, JS/TS, C/C++, and Java against hidden tests - a wedge into CS-department and technical-hiring evaluation none of the compared vendors compete in.
- Scorecard PDFs, bulk ZIP export, emailed results.

Inbuilt coding questions (execution flow)



The student never sees hidden-test inputs or expected outputs - only pass/fail after Submit.

WHY OUR RUNNING COST IS SO LOW

We removed the two biggest cost drivers of every other platform: server-side inference and continuous video.

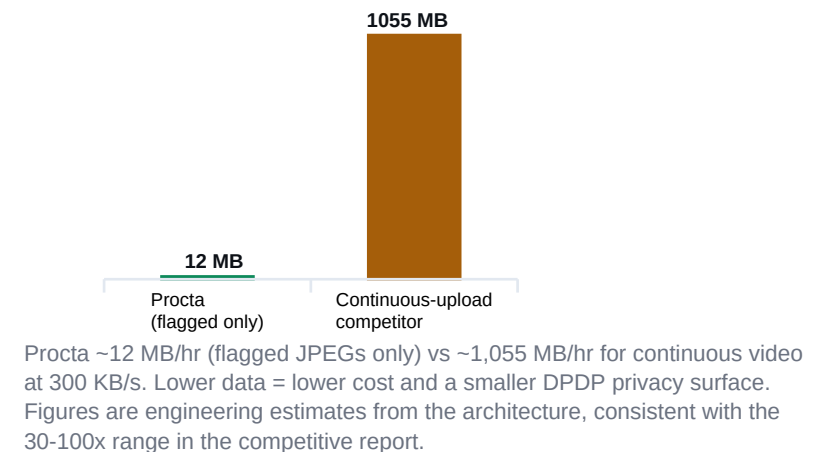
Most proctoring platforms pay twice per student: once to run ML on their own servers, and again to ingest and store continuous video. Procta does neither.

- **No server inference.** All ML runs on the student device, so cost does not scale with concurrent exam-takers - the expensive part is free to us.
- **No continuous video.** Only flagged JPEG snapshots are uploaded - roughly **88x** less data per exam-hour than continuous-upload competitors.
- **One Mumbai region.** A single Hostinger VPS plus Redis, not multi-region video-streaming infrastructure.

Net effect: the entire company runs on an 18-month budget of about Rs. 4.8 lakh - where a video-first competitor's cloud bill alone would exceed that.

The trade-off, stated plainly: on-device ML makes the desktop client heavy, not light - a one-time ~130 MB installer that caches roughly 2 GB of models locally and uses the student's CPU during the exam. That is the deliberate design choice, not a flaw: we shift the compute cost off our servers and onto hardware the student already owns, which is precisely what makes our operating cost and privacy surface so low. The CPU-adaptive hardware governor keeps it usable on budget laptops, and the install and model download happen once, not per exam.

Data uploaded per student per exam-hour



Privacy-by-construction, and DPDP Act alignment that shortens the IT security review.

Measure	How Procta implements it
Zero raw-video storage	Camera frames analysed on-device and discarded; only JPEG snapshots of flagged moments are uploaded as evidence.
On-device ML	RetinaFace, gaze, YOLOv8n, Silero VAD and Vosk all run client-side - no model inference or video on Procta servers.
DPDP Act alignment	Explicit pre-exam consent, data minimisation, 90-day default retention with auto-purge, Mumbai-region data residency, export and erasure endpoints.
Tenant isolation	Row-Level Security (native Postgres) on every table; JWT-scoped session ownership; evidence encrypted at rest in AWS S3 Mumbai; answer keys envelope-encrypted (AES-256-GCM) at the application layer.
Hardened client	Electron kiosk lockdown, sandboxed renderers, CSP, and runtime process-integrity scanning.
Transport and headers	HSTS (preload-eligible), CSP, ETag and GZip; Sentry error monitoring; structured audit logging.

Because Procta uploads far less personal data than continuous-video competitors, the data-minimisation and residency obligations under DPDP are easier to satisfy by construction - a procurement unlock for Indian institutions.

WHY CUSTOMERS CHOOSE PROCTA

The only vendor that is on-device, zero-raw-video, INR-priced, and publicly priced - all at once.

	Procta	US vendors	Mercer Mettl
On-device ML (no server inference)	Yes	No	No
Zero raw-video storage by default	Yes	No	No
Spoken-keyword detection (Hindi/English)	Yes	No	No
Phone-camera secondary view, no app install	Yes	Partial	Partial
INR pricing + GST + UPI Autopay	Yes	No	Partial
Public per-student rate card	Yes	No	No
Built for DPDP Act / Mumbai residency	Yes	Partial	Yes
Inbuilt coding-question judging	Yes	No	No
Live human-proctor team (where we are weaker)	No	Yes	Yes

Honest framing: incumbents still lead on live human-proctor headcount, deep Canvas/D2L LMS connectors, and assessment-authoring breadth. Procta competes on architecture, India fit, and price - not on enterprise feature depth. (Detail in the Competitive Comparison Report.)

MARKET TARGET

Land in the wedge that pays cash now: Indian coaching institutes.

Beachhead: coaching institutes

Allen, Aakash, Vedantu, PW and their peers run more proctored exams per month than most universities run per year, carry real edtech budgets, and have no loyalty to the incumbents. They are comfortable self-hosting and they pay in cash today.

Expand: universities and certification bodies

State universities and Tier-2/3 colleges follow on price and DPDP fit; AI short-answer grading is the wedge into larger institutions.

Budget figures are founder estimates from primary market conversations, not audited market research.

Cash now

coaching institutes buy today

DPDP

compliance is a procurement unlock

Go-to-market sequence

- 1. One coaching-chain pilot -> reference logo.
- 2. Per-student pricing + UPI Autopay self-serve loop.
- 3. Privacy/DPDP story to unlock universities.

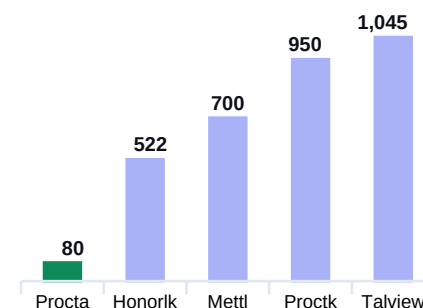
COMPANY COMPARISON

Same capability lens, honest about where we are weaker.

Capability	Procta	Mettl	Talview	Proctortrk	Honorlock
On-device ML inference	Y	N	N	N	N
Zero raw-video by default	Y	N	N	N	N
Spoken-keyword detection	Y	N	N	N	N
Phone-camera secondary view	Y	P	P	Y	N
UPI Autopay subscriptions	Y	N	N	N	N
Public per-student price card	Y	N	N	N	N
Live human-proctor escalation	N	Y	Y	Y	Y
Canvas LMS depth	P	Y	P	Y	Y
Inbuilt coding-question judging	Y	N	N	N	N

Y = advertised; P = partial; N = not advertised. Reused verbatim from the vetted Competitive Comparison Report.

Indicative price per attempt (INR)



Competitor figures are public-source midpoints converted at USD 1 = Rs. 95 (Mettl from Indian RFP disclosures). Indicative, not committed quotes - see the Competitive Comparison Report for ranges and sources.

THE ASK

Pre-seed ~Rs. 4.8 lakh for an 18-month ramped sprint.

Rs.4,81,900

Scenario B (ramped, 18 mo)

Rs.2,65,900

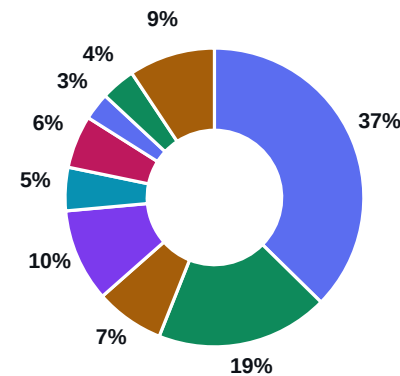
Scenario A (lean floor)

What the capital buys

- 18 months of operations with a real paid acquisition channel running.
- Rs. 10,000/month performance marketing for the full window.
- Prepaid 2-year hosting (primary + secondary server) and domain, dev tools, and Windows EV code-signing.
- Private Limited incorporation and its first 18 months of ROC/audit/CA compliance - a precondition of this round closing, not a deferred cost.

Excluded by design (so the rest is trusted): the second Windows cert issuance the 459-day validity cap forces around month 15, GST filing, founder living costs, Year-3 hosting renewal, and a future support hire.

Use of funds (Scenario B)



Performance marketing	Rs.1,80,000
Developer tools / software	Rs.90,000
Operations (email, Vercel)	Rs.36,000
Hosting and infrastructure	Rs.48,500
Setup and procurement (one-time)	Rs.22,600
Windows EV code-signing	Rs.27,550
Apple Developer Program	Rs.14,250
Pvt Ltd incorporation	Rs.18,000
Pvt Ltd compliance (18 mo)	Rs.45,000

Total Rs.4,81,900 - every category derives from the runway line items and reconciles exactly (no rounding residual). Detail in the Investment Requirement document.

WHY NOW

Working product, a market that pays cash, and a compliance moat that is opening right now.

- **Product is in active development** - 70+ features already built across proctoring, dashboard, grading and billing, with desktop clients for macOS and Windows; release-readiness work is ongoing.
- **DPDP Act is forcing the issue** - data minimisation and residency are now procurement criteria, and our architecture satisfies them by construction.
- **The cost base is the moat** - an order-of-magnitude price advantage that competitors cannot match without re-architecting away from server-side video.
- **The ask is small and the runway is long** - ~Rs. 4.8 lakh funds an 18-month sprint to the first named coaching-chain logo.

Companion documents: Procta_Investment_Requirement.pdf (capital detail), Procta_Competitive_Comparison_Report.pdf (sourced competitor analysis), Procta_Features.pdf (full feature surface).

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